

matmul_8bitraw_short — C = A × B on the T1 fabric (3×3 worked example)

Steps run top → bottom. Colour key: A by row; B by column-hue (so the V-load transpose turns columns into B^T rows); each Output cell keeps its final-position colour, so a result enters at column 0 and shifts right into place.

Example

0. Reference

C = A × B (true result)

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

B

10	11	12
13	14	15
16	17	18

Intermediate

Output C

84	90	96
201	216	231
318	342	366

Expected

Setup

1. Load H / V

vle8 v4,(A) [H]
csrwi 0x7c0,1
vle8 v5,(B) [V] → B^T

1	2	3
4	5	6
7	8	9

B^T

10	13	16
11	14	17
12	15	18

Iteration 0 (Full Steps)

2. k=2 Broadcast

csrwi 0x7c0,1
vrgather.vx v6,v5,2 [V]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

12	15	18
12	15	18
12	15	18

3. k=2 Multiply

csrwi 0x7c0,0
vmul.vv v6,v4,v6 [H]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

12	30	54
48	75	108
84	120	162

row-wise vmul.vv

4. k=2 Reduce

vredsum.vs v7,v6,v28 →
v7[0]=C*[2]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

12	30	54	Σ	96		
48	75	108	Σ	231		
84	120	162	Σ	366		

5. Shift →

vslideup.vi v8,v7,1
vmv.v.v v7,v8

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

→	96	
→	231	
→	366	

shift one column right →

SKIP 6. Broadcast, 7. k=1 Multitply

Iteration 1 (Simplified)

8. k=1 Reduce

vredsum.vs v7,v6,v28 →
v7[0]=C*[1]

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

11	28	51	Σ	90	96	
44	70	102	Σ	216	231	
77	112	153	Σ	342	366	

9. Shift →

vslideup.vi v8,v7,1
vmv.v.v v7,v8

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

→	90	96
→	216	231
→	342	366

shift one column right →

SKIP 10. Broadcast, 11. k=0 Multitply

Iteration 2 (Sim)

12. k=0 Reduce

vredsum.vs v7,v6,v28 →
full C
vse8 v7,(C)

1	2	3
4	5	6
7	8	9

10	13	16
11	14	17
12	15	18

10	26	48	Σ	84	90	96
40	65	96	Σ	201	216	231
70	104	144	Σ	318	342	366

Answer!!